

Marzan Samin Ashrafi

AI/ML Researcher | Software Quality Assurance Engineer

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Overview

Computer Science graduate and current MSc student at BRAC University with research and industry experience in Artificial Intelligence, Machine Learning, Software Quality Assurance, and Human-Computer Interaction. Published researcher with experience in developing and evaluating AI-driven solutions, with expertise in Explainable AI, Natural Language Processing, Software Testing, and Usability Evaluation.

Education

2015/01 – 2017/06 Dhaka, Bangladesh	Holy Cross Girls High School <i>Secondary School Certificate</i>
2017/07 – 2019/06 Dhaka, Bangladesh	Holy Cross College <i>Higher Secondary School Certificate</i>
2020/01 – 2024/03 Dhaka, Bangladesh	Military Institute of Science and Technology <i>Bachelor of Science in Computer Science and Engineering</i>
2026/02 – Present Dhaka, Bangladesh	BRAC University <i>Master of Science in Computer Science and Engineering</i>

Work Experience

2024/03 – 2025/09 Dhaka, Bangladesh	Dynamic Solution Innovators <i>Junior Software Quality Assurance Engineer</i>
2023/01 – 2023/03 Dhaka, Bangladesh	Code19 Bangladesh <i>Industrial Attachment</i>

Skills

Testing Skills

Test Plan and Test Case Writing, Software Testing, Manual Testing, API Testing, Automation Testing, Database Testing, Bug Reporting, QA Sign Off Report Writing, Product Requirement Writing

Database

MySQL, Oracle, PostgreSQL, MongoDB

Programming Languages

C, C++, Java, Dart, JavaScript, Python

Testing Tools

Jira, JMeter, Postman, Cypress

Tools & Platforms

Microsoft Office, Google Workspace, Google Colaboratory, Microsoft Teams, WinSCP, Datagrip, Android Studio, Figma, GitHub

Research

Artificial Intelligence, Machine Learning, Explainable AI (XAI), Natural Language Processing (NLP), Computer Vision, Usability Evaluation

Projects

A Knowledge-Guided Rule-Based Fairness Strategy Selection Framework for Mitigating Skin-Tone Bias in AI-Based Skin Disease Diagnosis

Research Project

Technology: Python, Machine Learning, Deep Learning, Computer Vision, Fairness Evaluation.

A research framework for identifying skin-tone, disease-class, and representation bias in AI-based skin disease diagnosis and recommending suitable fairness mitigation strategies based on dataset characteristics.

An Explainable Decision Support System for Medical Spam Detection Using Ensemble Learning and Fuzzy TOPSIS

AI/ML and Explainable AI Project

Technology: Python, NLP, TF-IDF, Scikit-learn, Ensemble Learning, Fuzzy TOPSIS, Explainable AI.

An explainable hybrid framework for detecting medical spam and misinformation by combining TF-IDF-based text classification, Naive Bayes, Logistic Regression, and SVM with ensemble probability aggregation, fuzzy semantic risk calibration, and XAI-based reasoning.

Fake Logo Detection Using Deep Learning

Pattern Recognition Project

Technology: Deep Learning, Python, TensorFlow, OpenCV.

A system to differentiate between the real logo and fake logo of a brand from an image using a convolutional neural network.

A Novel Hand-Free Interaction Technique Based on The Nose-Teeth Movement

Thesis Project

Technology: Machine Learning, Python, PyQt5, Mediapipe's Facemesh for Face and Facial Point Recognition.

A system made to enable people with physical limitations to use desktop computers without assistance from others.

A Study to Find Out a User-Friendly AI Toolkit by Evaluating Usability of AutoML Tools

Human-Computer Interaction Project

Technology: LaTeX, AutoML Tools.

A study evaluating the usability of three AutoML tools: Akkio, H2O, and Amazon SageMaker Canvas.

E-Incubator: IoT Based Automated Egg Incubator System

Integrated Design Project

Technology: Figma, Flutter, Android Studio, Dart, Realtime Database, Firestore Database, Firebase Functions, Firebase Authentication, Arduino.

An egg incubator system that enhances incubation conditions and advances the hatching process.

Conferences

IoT Enabled Smart Egg-incubator with Remote Monitoring and Control System

3rd International Conference on Computing Advancements (ICCA 2024) - Second Author

Citation: 03

Assessing Usability of AutoML Tools: A Multi-Method Study

2024 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE) - First Author